Exploration of Homelessness in the United States: A Data-Driven Analysis

Paul Zee-Cheng

Group Section 1. Github User Name PaulZUC.

The following visualiztions and analyses explore the issue of homelessness in the United States, utilizing data from the ASC, PIT Counts, the Urban Institute's Mobility Initiative, and the Census Opportunity Atlasto provide insights into the demographics, trends, and factors contributing to homelessness. The goal is to identify patterns and inform potential policy interventions to address this critical social issue.

This exploration faced many difficult issues, not the least of which was lack of longitudinal data to explore what is essentiall a longitudinal analysis of poverty.

The motivating question was inspired by a discusion of the "Kill Line" which is itself the idea that there exists an economic "point of no return" for American workers, where an economic shock or hardship will reduce them to a point of poverty that they cannot recover from.

To this end, after initial exploration, an as-yet-unpublished working paper from Bruce Meyers et al, "Homelessness and the Persistence of Deprivation: Income, Employment, and Safety Net Participation" indicates that this is not the case.

Rather, this paper demonstrates through select longitudinal panel data of the same individuals over time that permanent, rather than transitional or temporary economic shocks, is the greatest indicator of homelessness. Variables such as Social Security Insurance and employment persist through homelessness. This in turn indicates that the homless population is not impoverished because of a lack of government benefits or employment, but in addition to these.

The dashboard created thus attempts to explore the relationship between homelessness, unemployment, and government assistance over time, across all states, over a change of a 10 year period.

For further explanation of the data, please see presentation slide deck in this same folder.

Figure 1 shows the trends in living standards over time.

Figure 2 shows the trends in living standards over time.

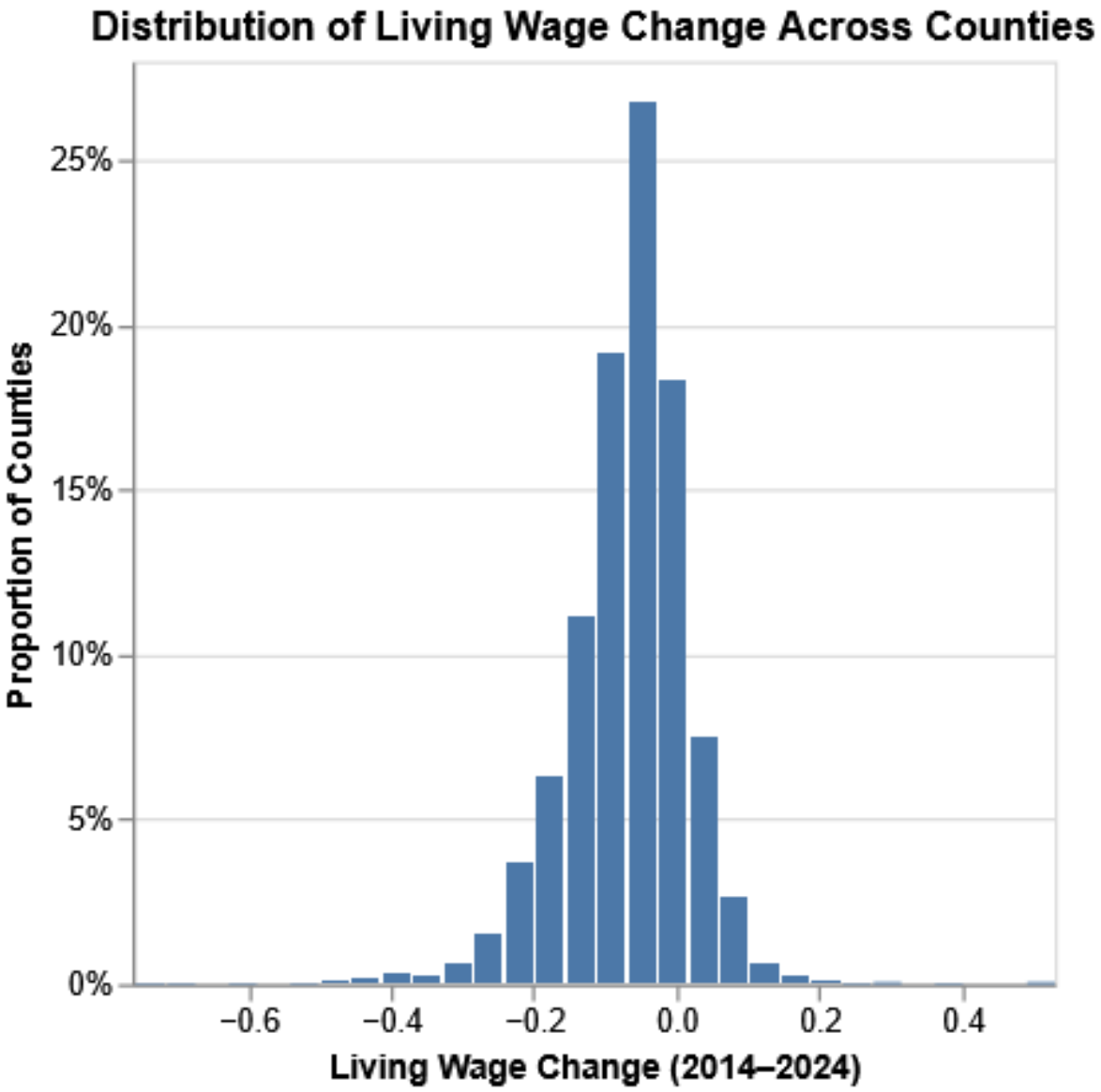


Figure 1: Cost of Living changes by county, over 10 years.

Figure 3 shows the trends in living standards over time.

Figure 4 shows relationship between homelessness, unemployment, and government assistance for all states over a 10 year period, as visualized in a locally deployed streamlit app.

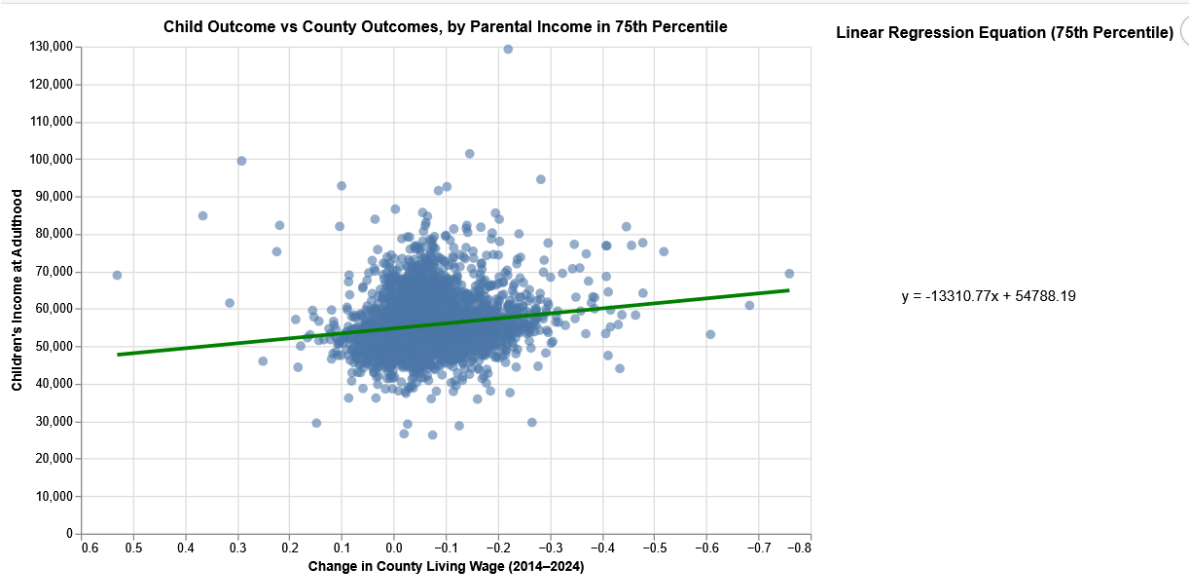


Figure 2: Adult earnings of children who's parents were in the top 75th percentil of income, versus change in living standards of county of children during development.

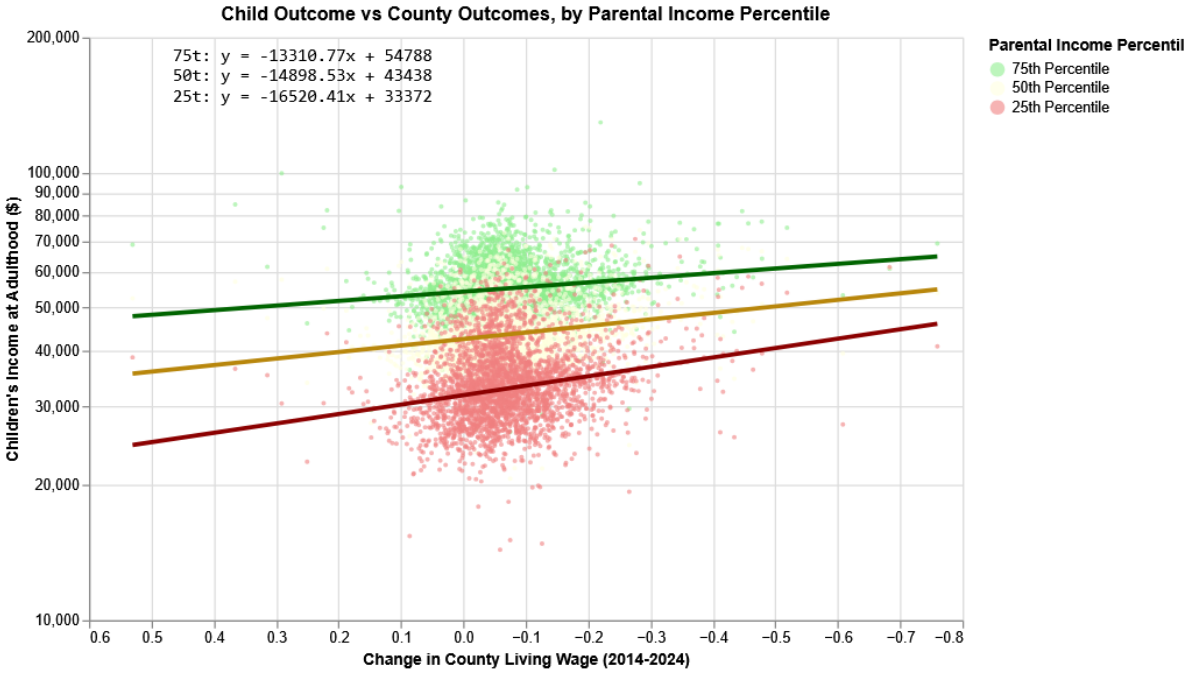
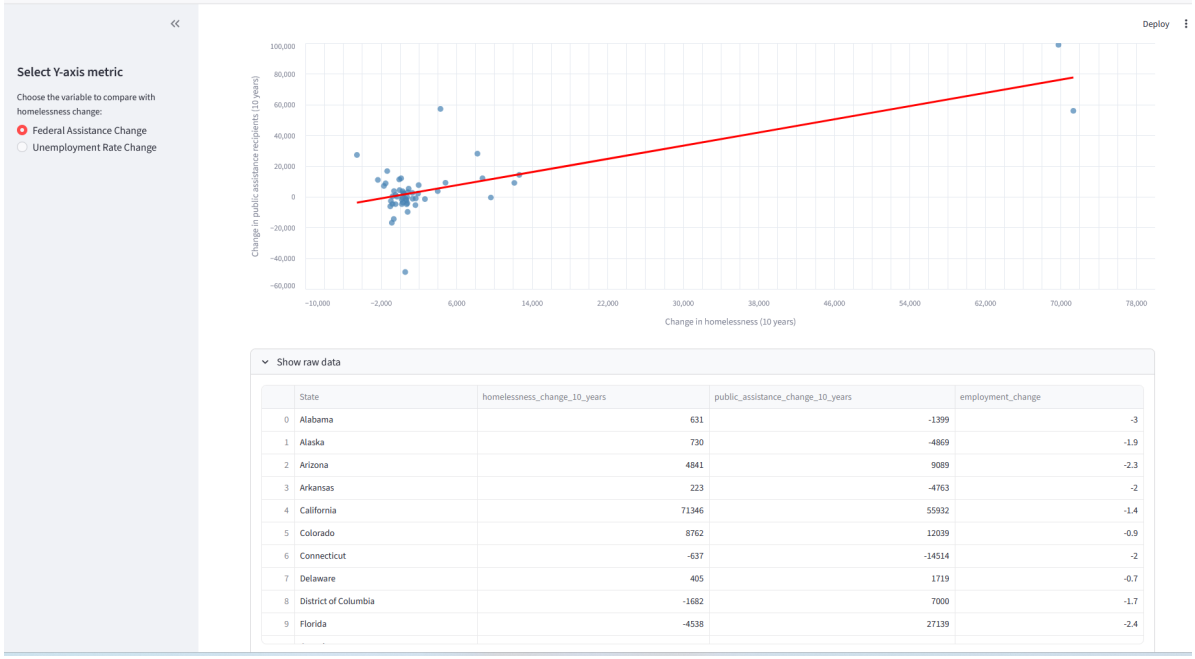


Figure 3: Adult earnings of children who's versus change in living standards of county of children during development, separated by income quartile earnings of parents, for 75th, 50th, and 25th quartile.



	State	homelessness_change_10_years	public_assistance_change_10_years	employment_change
0	Alabama	631	-1399	-3
1	Alaska	730	-4869	-1.9
2	Arizona	4841	9089	-2.3
3	Arkansas	223	-4763	-2
4	California	71346	55932	-1.4
5	Colorado	8762	12039	-0.9
6	Connecticut	-637	-14514	-2
7	Delaware	405	1719	-0.7
8	District of Columbia	-1682	7000	-1.7
9	Florida	-4538	27139	-2.4

Figure 4: Sample of locally deployed streamlit app, saved as app.py, tracking relationship between homelessnes, unemployment, and government assistance for all states over a 10 year period.